

Firebase

Login, database and hosting without your own server · Equigrow short notes

Setup (web, modular SDK)

```
import { initializeApp } from "firebase/app";
import { getAuth, signInWithEmailAndPassword } from "firebase/auth";
import { getFirestore, doc, getDoc, setDoc } from "firebase/firestore";

const app = initializeApp(firebaseConfig);
const auth = getAuth(app), db = getFirestore(app);
```

Authentication

```
await signInWithEmailAndPassword(auth, email, password);
onAuthStateChanged(auth, user => { /* user is null when signed out */ });
```

Firestore

```
await setDoc(doc(db, "users", uid), { name: "Asha", branch: "CSE" });
const snap = await getDoc(doc(db, "users", uid));
if (snap.exists()) console.log(snap.data());
```

Data lives in **collections** of **documents** (JSON-like). Arrays inside arrays are not allowed.

Security rules

```
match /users/{userId} {
  allow read, write: if request.auth != null && request.auth.uid == userId;
}
```

Never leave rules open (allow read, write: if true). Rules are what keep your users' data private.